

Problem K: Kid's Spiral Problem

Time Limit: 5 seconds

Description

2	17	16	15	14	13
1	18	5	4	3	12
0	19	6	1	2	11
-1	20	7	8	9	10
-2	21	22	23	24	25
	-2	-1	0	1	2

A spiral on a grid of size $(2n+1) \times (2n+1)$ has been constructed as follows. Number 1 is in the center square, number 2 is to the right of it, and then we continue place the positive integers in order along the spiral in counterclockwise fashion. Now, given 2 coordinates indicating 2 corners of a rectangle, find the sum of all numbers in the enclosing rectangle. See the figure above for example.

Input

A number of of inputs (≤ 100), each starting with line contains two integers and n ($1 \leq n \leq 10^9$) and q ($1 \leq q \leq 100$): the size of the grid and the number of queries. After this, there are lines, each containing four integers (x_1, y_1) and (x_2, y_2) in that order, where $-n \leq x_1, y_1, x_2, y_2 \leq n$. This is the 2 corners of the rectangle.

Output

For each input, output the answer modulo **1000000007**.

Sample Input

```
2 3
0 -2 1 1
-1 0 1 0
1 2 1 2
```

Sample Output

```
74
9
14
```